

# **L6a:** Metabolic Engineering and Flux Balance Analysis

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Stoichiometry, reaction bounds, and production limits  
CHEME 5800 Cornell University

# Objectives for Today

How do material balances and biological constraints limit the production of a desired compound?

## Today's concepts

- You will **represent a metabolic reaction network as a stoichiometric matrix** and interpret its rows, columns, and coefficient signs.
- You will **formulate the flux balance analysis problem**, a linear program for the unknown fluxes, and explain its objective and the constraints that define the feasible solution space.
- You will **develop and interpret flux bounds** from reaction reversibility and enzyme capacity, and explain their biological meaning.

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**Lecture:** [Metabolic Engineering and Flux Balance Analysis](#)

# Lecture and Examples

## **Lecture:** [Metabolic Engineering and Flux Balance Analysis](#)

In this lecture, we introduce flux balance analysis (FBA), a tool based on linear programming, to estimate reaction rates (called fluxes) in a metabolic network.

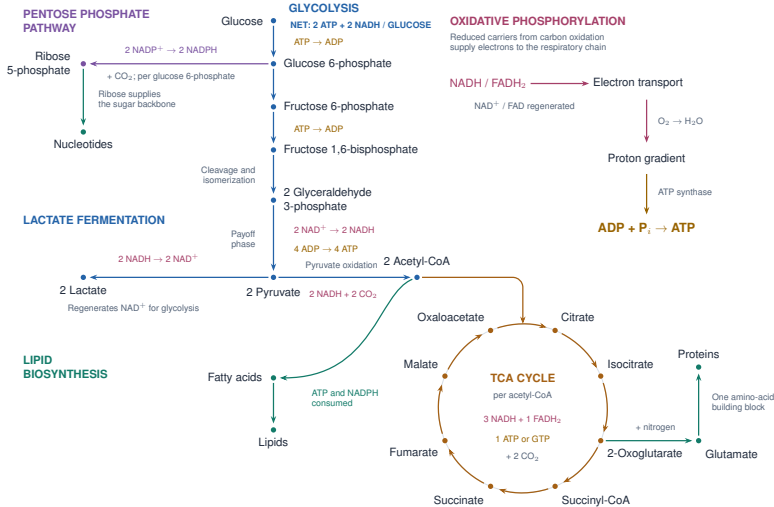
## **Example:** [Calculate fluxes in a urea-cycle model](#)

In this example, we build a stoichiometric model of the urea cycle in human cells, estimate reaction directions and enzyme capacities to establish the flux bounds, and solve a linear program to maximize urea export. The same approach applies to any metabolic network.

We can explore the connections among metabolic pathways in the [KEGG metabolic pathways map](#).

# Central metabolism

Carbon flow, energy production, and biosynthetic branches



Selected pathways; arrows may combine reactions. Glycolysis and pyruvate-oxidation yields are per glucose; TCA yields are per acetyl-CoA. Cofactors shown are selective. Compartments, complete stoichiometry, and alternative routes are omitted; pathway availability depends on the organism.

# What Is Metabolic Engineering?

Metabolic pathways encode the enzyme-catalyzed reactions that convert nutrients into biomass and energy. Some pathways are highly conserved across species, such as **central metabolism**: glycolysis, the pentose phosphate pathway, and the TCA cycle.

**Metabolic engineering** is the directed modification of an organism's metabolic pathways through genetic and regulatory changes to increase production of a desired compound or to enable new products.

The engineering question is: how can we redirect carbon, nitrogen, and energy toward a target molecule while meeting the cell's other requirements? We describe the flow of material using **metabolic fluxes**: reaction rates per unit volume or biomass.

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**Early work:** [Bailey \(1991\)](#); [Stephanopoulos and Vallino \(1991\)](#).

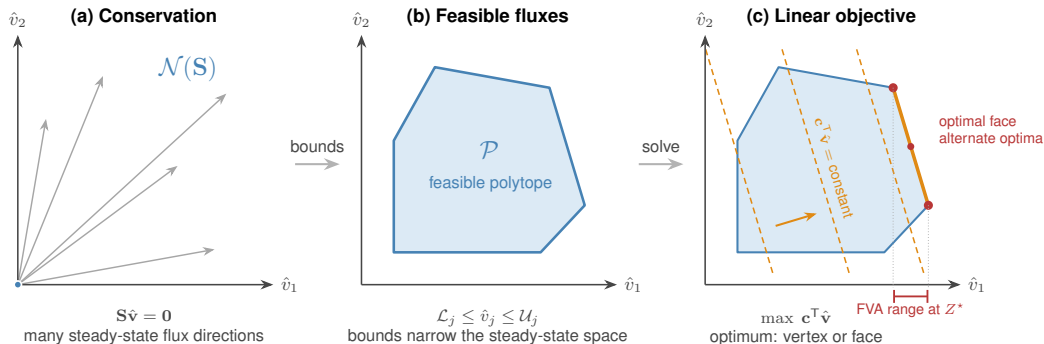
# Flux Balance Analysis

Flux balance analysis (FBA) estimates optimal metabolic reaction rates (fluxes) from material balances and biological constraints.

FBA uses linear programming to find the flux distribution that optimizes a chosen objective function, subject to stoichiometric and capacity constraints, at a pseudo-steady state.

1. The stoichiometric matrix encodes the reaction network and the steady-state material balances.
2. The flux bounds restrict each reaction rate using reversibility and enzyme capacity.
3. The linear program combines the balances, bounds, and an objective, such as maximizing product export.

# Conservation, Bounds, and Alternate Optima



Balances define the null space, bounds cut it to the feasible set  $\mathcal{P}$ , and the objective selects an edge, so the optimum is **not unique**. **Flux variability analysis** finds the range of each flux at  $Z^*$ .

# Stoichiometric Matrix

We build  $\mathbf{S}$  from a **metabolic reconstruction**: a curated set of reactions, metabolites, and stoichiometric coefficients. Let  $\mathcal{M}$  be the set of modeled species and  $\mathcal{R}$  the set of reactions. The entry  $\sigma_{ij}$  is the net coefficient of species  $i$  in reaction  $j$ :

$$\mathbf{S} = [\sigma_{ij}] \in \mathbb{R}^{|\mathcal{M}| \times |\mathcal{R}|}.$$

- One **column** describes a reaction; one **row** collects contributions to a species balance.
- $\sigma_{ij} > 0$ : species  $i$  is a **product** of reaction  $j$ .  $\sigma_{ij} < 0$ : species  $i$  is a **reactant**.
- $\sigma_{ij} = 0$ : species  $i$  is not involved, or the net stoichiometry is zero.

The matrix records net stoichiometry. Regulation enters through the flux bounds, not the stoichiometric matrix.



# Biological Data Resources

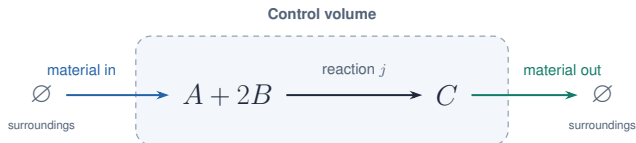
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| Resource                    | What we use it for                                                                |
|-----------------------------|-----------------------------------------------------------------------------------|
| <a href="#">KEGG</a>        | We examine pathways and connections among genes, enzymes, and reactions.          |
| <a href="#">BioCyc</a>      | We find pathway and genome information for a particular organism.                 |
| <a href="#">BiGG Models</a> | We obtain metabolic models with standardized metabolite and reaction identifiers. |
| <a href="#">BRENDA</a>      | We find enzyme properties and kinetic measurements to estimate flux bounds.       |
| <a href="#">BioNumbers</a>  | We find measured biological quantities and their literature sources.              |

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**Readings:** Resource names link to the papers listed in the lecture.

# Example: One Reaction Gives One Column



The dashed boundary encloses the modeled species; material enters and leaves through it. Reaction  $j$  consumes one unit of  $A$  and two units of  $B$  to produce one unit of  $C$ . With rows ordered  $(A, B, C)$ :

$$A + 2B \longrightarrow C, \quad \mathbf{s}_j = \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix}, \quad \mathbf{s}_j \hat{v}_j = \begin{bmatrix} -\hat{v}_j \\ -2\hat{v}_j \\ \hat{v}_j \end{bmatrix}.$$

The coefficient  $-2$  means  $B$  is consumed at twice the reaction flux. A negative flux reverses the signs; the column stays the same.

# Exchange Reactions Cross the Boundary

An **exchange reaction** connects a metabolite to the surroundings, written  $\emptyset$ , which we do not balance. Its column has a single nonzero entry. For the reactant  $A$  supplied by exchange reaction  $k$ :

$$\emptyset \longrightarrow A, \quad \mathbf{s}_k = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

A positive flux brings  $A$  into the control volume; a negative flux removes it. The urea example writes every exchange this way. The BiGG models in the L6b lab write  $A \longrightarrow \emptyset$ , so uptake is negative there.

The sign convention does not affect the FBA solution, but it does affect the interpretation of the fluxes.

# A Model for Flux Bounds

Flux bounds are **integrative**: they bring many types of genetic and biochemical information into the problem. For enzyme-catalyzed reaction  $j$ , a general model for the bounds is:

$$\underbrace{-\delta_j \left[ \overbrace{V_{max,j}^\circ \left( \frac{e}{e^\circ} \right) \theta_j(\dots) f_j(\dots)}^{\text{reverse: other functions or parameters?}} \right]}_{\mathcal{L}_j} \leq \hat{v}_j \leq \underbrace{V_{max,j}^\circ \left( \frac{e}{e^\circ} \right) \theta_j(\dots) f_j(\dots)}_{\mathcal{U}_j}.$$

Here  $V_{max,j}^\circ$  is the maximum reaction velocity at a characteristic enzyme abundance,  $e/e^\circ$  corrects for the actual abundance,  $\theta_j$  is the allosteric activity,  $f_j$  the substrate saturation, and  $\delta_j$  the reversibility parameter.

This model assumes equal forward and reverse capacities for a reversible reaction. We fix every quantity **before solving** the linear program.

# Reference Enzyme Capacity

Let  $k_{cat,j}^\circ$  be the reference catalytic turnover number and  $e^\circ$  the characteristic enzyme abundance. The reference capacity is:

$$V_{max,j}^\circ = k_{cat,j}^\circ e^\circ.$$

| Quantity          | Meaning                                                  | Units                                |
|-------------------|----------------------------------------------------------|--------------------------------------|
| $V_{max,j}^\circ$ | This is the maximum rate at reference enzyme abundance.  | $\text{mmol gDW}^{-1} \text{h}^{-1}$ |
| $k_{cat,j}^\circ$ | This is the turnover rate at substrate saturation.       | $\text{h}^{-1}$                      |
| $e^\circ, e$      | $e^\circ$ and $e$ denote reference and actual abundance. | $\text{mmol gDW}^{-1}$               |

The ratio  $e/e^\circ$  scales the reference capacity to the actual abundance of the enzyme catalyzing reaction  $j$ .

# Activity, Saturation, and Reversibility

The activity factor  $\theta_j$ , saturation factor  $f_j$ , and reversibility parameter  $\delta_j$  are dimensionless:

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| Quantity                     | Meaning                                                                                                  |
|------------------------------|----------------------------------------------------------------------------------------------------------|
| $\theta_j(\dots) \in [0, 1]$ | The activity factor gives the fraction of maximal activity available under allosteric regulation.        |
| $f_j(\dots) \in [0, 1]$      | The saturation factor describes substrate availability. A value of one represents saturating substrates. |
| $\delta_j \in \{0, 1\}$      | The reversibility parameter is zero for a forward-only reaction and one when reverse flux is allowed.    |

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Activity and substrate availability can reduce the capacity below  $V_{max,j}^\circ(e/e^\circ)$ .

# Simplified Bounds Model

Let's assume  $e/e^\circ \sim 1$ , no allosteric inputs  $\theta_j(\dots) \sim 1$ , and saturating substrates  $f_j(\dots) \sim 1$ . Then the flux bounds are:

$$\underbrace{-\delta_j V_{max,j}^\circ}_{\mathcal{L}_j} \leq \hat{v}_j \leq \underbrace{V_{max,j}^\circ}_{\mathcal{U}_j}.$$

- For a **forward-only reaction**,  $\delta_j = 0$ , so the flux satisfies  $0 \leq \hat{v}_j \leq V_{max,j}^\circ$ .
- For a **reversible reaction**,  $\delta_j = 1$ , so the flux satisfies  $-V_{max,j}^\circ \leq \hat{v}_j \leq V_{max,j}^\circ$ .

The bounds are a function of the maximum reaction velocity, the turnover number, and our assumed characteristic enzyme abundance.

# Exchange Bounds Specify the Surroundings

Exchange bounds limit nutrient uptake and product secretion. In the urea example, we write every exchange as uptake from the surroundings, so **positive flux represents uptake** and negative flux represents export.

- The upper bound limits the uptake rate.
- An upper bound of zero prevents uptake through that reaction.
- A negative lower bound sets the largest allowed export rate.
- A lower bound of zero prevents export through that reaction.

The exchange bounds specify which nutrients the system can take up and which products it can release. Internal reactions have separate bounds based on enzyme capacity and reaction direction.



# The Flux Balance Approximation

The FBA problem has the same structure as the minimum-cost flow problem:  $\mathbf{S}$  plays the role of the incidence matrix, and the exchange reactions are the sources and sinks.

The material balances rest on two assumptions:

- The amount of each metabolite per gram of biomass stays constant.
- Growth dilution is small compared with production and consumption.

Then, for each species  $i \in \mathcal{M}$ :

$$\sum_{j \in \mathcal{R}} \sigma_{ij} \hat{v}_j = 0, \quad \text{or} \quad \mathbf{S} \hat{\mathbf{v}} = \mathbf{0}.$$

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**Derivation:** [Flux balance analysis material balances](#)

# Flux Balance Analysis Problem

Let  $c_j$  be the objective weight of reaction  $j$ , and let  $\mathcal{L}_j$  and  $\mathcal{U}_j$  be its flux bounds. With the coefficients, weights, and bounds fixed, we solve:

$$\begin{aligned} & \underset{\hat{v}_1, \hat{v}_2, \dots, \hat{v}_{|\mathcal{R}|}}{\text{maximize}} && \sum_{j \in \mathcal{R}} c_j \hat{v}_j \\ & \text{subject to} && \sum_{j \in \mathcal{R}} \sigma_{ij} \hat{v}_j = 0, \quad i \in \mathcal{M} \quad (\text{material balance constraints}) \\ & && \mathcal{L}_j \leq \hat{v}_j \leq \mathcal{U}_j, \quad j \in \mathcal{R} \quad (\text{thermodynamics and kinetic constraints}) \end{aligned}$$

A **feasible** distribution satisfies all balances and bounds. The objective selects an **optimal** distribution from the feasible set; the optimal solution may **not be unique**.

# An Objective for Product Export

Let  $p$  denote the urea exchange reaction, b4. We write this reaction as uptake from the surroundings, so **negative flux represents export**. We set its objective weight to  $-1$  and all other weights to zero:

$$c_j = \begin{cases} -1, & j = p, \\ 0, & j \neq p, \end{cases} \quad \sum_{j \in \mathcal{R}} c_j \hat{v}_j = -\hat{v}_p.$$

Maximizing  $-\hat{v}_p$  makes the urea exchange flux as negative as the balances and bounds allow, which maximizes urea export.

We express metabolic fluxes in  $\text{mmol gDW}^{-1} \text{h}^{-1}$ , where gDW means grams of dry biomass. Each bound has the units of its flux.

# Example: Calculate Fluxes in a Urea-Cycle Model

In this example, we calculate the maximum urea export rate for a urea-cycle network in human cells.

1. We build the stoichiometric matrix, species and reaction lists, and default bounds from the network file.
2. We assign reaction directions from Gibbs energy estimates and set enzyme capacities from turnover numbers and a reference enzyme abundance.
3. We solve the linear program, inspect the optimal fluxes, and check the solution against the balances and bounds.

The same approach applies to any metabolic network, and the bounds show how many types of biochemical information integrate into one problem.

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**Example:** [Calculate fluxes in a urea-cycle model](#)

# Summary

We formulated flux balance analysis as a linear program, combining a metabolic network, biological constraints, and a production objective.

## Key takeaways

- **Stoichiometric matrix.** Rows are metabolites, columns are reactions, entries are net stoichiometric coefficients.
- **Flux balance analysis.** A linear program for the unknown fluxes: a production objective, subject to balances and bounds.
- **Flux bounds.** Turnover number times enzyme abundance; they integrate biochemical information into the problem.

FBA lets us predict how network changes affect production of a desired compound.